PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM IV TAHOSSA NORTH WESTERN ZONE MOCK EXAMINATION 2025

036 INFORMATION AND COMPUTER STUDIES

MARKING SCHEME

I. D
II. C
III. B
IV. B
V. D
VI. A
VII. A
VIII. D
IX.A
X. D

QUESTION 2

Column A	**I**	**Ii**	**Iii**	**Iv**	**V**	**Vi**
Column B	**G**	**F**	**E**	**C**	**B**	**A**

3

I	Slow Computer Performance	
	The slow performance could be due to multiple factors such as insufficient RAM, too many background applications running, malware, or a fragmented hard drive **(1 mark)**	2 marks
	To resolve this, you could close unnecessary applications, run a disk cleanup, scan for malware, upgrade RAM if possible, or consider upgrading to an SSD for faster performance. **(1 mark)**	
Ii	-Printer Not Responding	2 marks
	The printer may not be responding due to connectivity issues, such as being offline, problems with the driver, or a stuck print queue. **(1 mark)**	
	To troubleshoot, check the printer's status, ensure it is online, restart both the printer and the computer, clear the print queue, and reinstall or update the printer drivers **(1**	

	mark)	
Iii	Internet Connectivity Issue	
	The connectivity issue could be due to incorrect network settings, a disabled network adapter, or issues with the router or modem**. (1 mark)**	2 marks
	Try restarting the computer and router, checking network settings, running the built-in Network Troubleshooter, and ensuring the network adapter is enabled **(1 mark)**	
(iv)	Software Crashes	2 marks
	Software crashes could be caused by bugs in the application, compatibility issues, or insufficient system resources. **(1 mark)**	
	To troubleshoot, check for software updates, ensure compatibility with the operating system, or consider reinstalling the application. Monitoring system resources while running the application can also be helpful**. (1 mark)**	

4. ANSWERS

(a) Reasons to Use MS Word for Writing a Report Instead of a Typewriter:

Editing and Formatting Flexibility

Spell Check and Grammar Tools

Inserting Images and Tables

(b) Three Common Features of a Word Processor:

Text Formatting

Spell Check

Page Layout and Design

(c) Reasons Why a Desktop Publisher is Preferred Over a Word Processor for Designing Documents:

Advanced Layout and Design Tools

Better Control Over Typography.

High-Quality Graphics and Image Integration.

Multiple Page Layouts and Templates.

Professional output quality

ANSWERS QUESTION NO 5

		FUNCTION	Marks
	A-INSULATION (MARK)	-It used to maintain the spacing between the center conductor and shield (MARK)	**1MARK**
	B-COPPER MESH (MARK)	-It serve as grounding shield. - it prevents signals in the cable from leaking out (MARK)	**1 MARKS**
	C- COPPER WIRE (MARK)	Carries electrical signals (data) transmitted through the cable. (MARK)	**1 MARKS**
	D-OUT SIDE INSULATION (MARK)	Protect the cable through physical damage, moisture and environmental factors. (MARK)	**1 MARKS**
	-high bandwidth: coaxial cables can support higher bandwidths compared to twisted cables allowing for transmission of more data at a faster speed -better shielding: the design of coaxial cable, with outer conductor provides better shielding aginst electromagnetic interference (EMI) and radio frequency interference (RFI) - longer distance transmission: coaxial cables can transmit signals over longer distances without significance signal loss compared to twisted		**1 MARKS**
	Application of coaxial cables :cable television and high speed internet		**1 MARKS**
(b)	(i) STAR TOPOLGY (diagram)		**2 MARK**
	- Ease of installation and management MARK) The failure of one computer it affect only that computer MARK)		**1 MARKS**

6.**Question 6:**

(20 Marks)

Answer the following questions about spreadsheets:

(a)(5marks)
Define what a spreadsheet is and explain the main components of a typical spreadsheet application.

A spreadsheet is a digital tool used for organizing, analyzing, and storing data in a tabular format. It consists of rows and columns where users can input data, perform calculations, and visualize information. Spreadsheets are widely used in business, finance, data analysis, and various other fields due to their versatility.

Main Components of a Typical Spreadsheet Application:

1. Cells: The basic unit of a spreadsheet where data is entered. Each cell is identified by a unique address, combining its column (letter) and row (number), e.g., A1, B3.

2. Rows and Columns: Rows run horizontally and are numbered (1, 2, 3, …). Columns run vertically and are labeled with letters (A, B, C, …). The intersection of a row and column forms a cell.

3. Worksheet: A single page or tab in a spreadsheet that holds the rows and columns. A typical spreadsheet may contain multiple worksheets, which can be accessed via tabs at the bottom.

4. Formula Bar: Located above the spreadsheet grid, the formula bar displays the contents or formulas of the currently selected cell. It's where you can enter or edit data and formulas.

5. Formulas and Functions: Formulas are mathematical expressions (e.g., `=A1+B1`) used to calculate results based on the data in the cells. Functions are predefined formulas (e.g., `SUM()`, `AVERAGE()`) that perform common calculations.

6. Charts and Graphs: Spreadsheets allow users to create visual representations of data (e.g., bar charts, line graphs) to better understand and present data trends.

7. Data Formatting: Cells can be formatted for specific data types (e.g., currency, date, percentage) and styles (e.g., font size, cell color, borders) to enhance readability.

8. Headers: Row headers are the numbered rows on the left, while column headers are the lettered columns at the top. These help identify the position of each cell in the spreadsheet.

9. Toolbar and Menus: Spreadsheet applications come with a toolbar and menus that offer various functions for formatting, data manipulation, and analysis (e.g., sorting, filtering, conditional formatting).

(b)(5marks)
Explain the difference between relative, absolute, and mixed cell references in Excel. Provide examples for each.

1. Relative Cell Reference

A relative cell reference changes when you copy the formula to another location. The reference is relative to the position where the formula is copied.

Example:
Formula in cell B2: =A1
If you copy this formula to cell B3, it will change to =A2, because the reference is relative to the formula's original position.

2. Absolute Cell Reference

An absolute cell reference remains constant no matter where you copy the formula. It is denoted by dollar signs ($), and it locks both the row and the column.

Example:
Formula in cell B2: =A1
If you copy this formula to cell B3, it will still reference cell A1 (i.e., it doesn't change).

3. Mixed Cell Reference

A mixed cell reference locks either the row or the column, but not both. It is a combination of relative and absolute references.

Example 1 (Locking the row):
Formula in cell B2: =A$1

If you copy this formula to cell B3, the column reference A will change relative to the new position, but the row reference $1 will stay fixed.

Example 2 (Locking the column):
Formula in cell B2: =$A1
If you copy this formula to cell C2, the row reference 1 will change, but the column reference $A will remain fixed.

(c)(5marks)
Here are the purposes of the functions in a spreadsheet:

SUM: Adds up a range of numbers or values. For example, =SUM(A1:A5) would total the values in cells A1 through A5.

AVERAGE: Calculates the average (mean) of a set of numbers. For example, =AVERAGE(B1:B5) would return the average of the values in cells B1 through B5.

VLOOKUP: Searches for a value in the first column of a range and returns a corresponding value from a specified column in the same row. For example, =VLOOKUP("Apple", A2:B10, 2, FALSE) would search for "Apple" in the first column (A) and return the value in the second column (B) of the same row.

COUNTIF: Counts the number of cells in a range that meet a specific condition. For example, =COUNTIF(C1:C10, "Yes") would count how many cells in the range C1 to C10 contain the value "Yes".

IF: Evaluates a condition and returns one value if the condition is true and another if it is false. For example, =IF(D1>100, "Over 100", "Under 100") would return "Over 100" if the value in D1 is greater than 100 and "Under 100" otherwise.

(d)(5marks)
What is conditional formatting in spreadsheets? Explain how it can be applied to highlight cells based on certain criteria. Provide two examples where conditional formatting would be useful.

Conditional formatting in spreadsheets is a feature that allows you to apply different formatting styles (such as colors, fonts, or cell borders) to cells based on specific conditions or criteria. This makes it easier to visually identify trends, outliers, or patterns within your data. When a condition is met, the cell's formatting changes automatically.

How to Apply Conditional Formatting:

1. Select the Range of Cells: Highlight the cells you want to apply the conditional formatting to.

2. Open the Conditional Formatting Menu: In Excel or Google Sheets, navigate to the “Format” or “Conditional Formatting” option.

3. Define the Condition: Choose the condition you want to apply (e.g., cells greater than a certain value, cells containing specific text, or cells based on a formula).

4. Choose the Formatting Style: Select the format (e.g., fill color, text color) you want to apply when the condition is met.

5. Apply the Rule: Once the rule is defined, the formatting will automatically adjust based on the data.

Two Examples Where Conditional Formatting Is Useful:

1. Highlighting Sales Above Target

2. Identifying Expired Items in Inventory:

7ANSWERS

(a) **Program to use** – A web-based email service like **Gmail, Yahoo Mail, or Outlook**.
(b) **Steps to create an email account**

Open a web browser and go to the email provider's website (e.g., www.gmail.com).

Click on **"Create Account"** or **"Sign Up"**.

Enter required details like name, desired email address, and password.

Provide a phone number for verification.

Verify the account using a code sent via SMS.

Complete security questions and accept the terms & conditions.

Click **"Finish"** and access the inbox.

(c) **Explanation of Spam, CC, and BCC**

Spam – Unwanted or junk emails, often advertisements or phishing attempts.

Carbon Copy (CC) – Sends a copy of an email to multiple recipients, and all recipients can see who else received the email.

Blind Carbon Copy (BCC) – Sends a copy of an email to multiple recipients, but their addresses remain hidden from each other.

8. (a) Examples of Desktop Publishing (DTP) software are:

- Microsoft (MS) publisher,
- PageMaker
- Express publisher
- Ventura publisher
- Adobe Creative Suite and
- QuarkXPress.

(01.5 Marks @any two examples of Desktop Publishing (DTP) software, total 03Marks)

(b) The differences between Desktop Publishing and Microsoft Word are:

(i)**DTP Software** is designed for creating professional level publications such brochures,

magazines, newspapers, and other visually complex documents ***WHILE*** **MS word** is word

processing tools which is used for creating text-heavy documents like letters, reports, essays

and basic newspapers, focusing more on text formatting and documents editing.

(ii) **DTP** offers advanced tools for controlling layouts, precise placement of text and images,
and advanced typography options ***WHILE*** **MS Word** offers basic layout and formatting tools
such as margins, text alignment, and simple image insertion, but it is not as flexible or
precise as DTP tool for complex layouts.

(iii) **DTP software** integrates well with graphics and allows for sophisticated images handling,
vector graphics and high quality output, ***WHILE*** **MS Word** does not allows inserting images,
charts, and tables, it is not designed for high quality image manipulation or professional
graphics work.

(iv) **DTP Software** is optimized for professional printing, offering high quality output for offset
printing and detailed color management. It supports various formats like PDF which are
suitable for printing, ***WHILE*** **MS Word** printing capabilities are more basic and suited for
standard office documents. It does not offer advanced printing features like color separation
or advanced PDF options for professional printing.

(v) **DTP Software** typically have more complex interface with various panels for controlling
typography, layouts grids, color palettes and image placement, ***WHILE*** **MS Word software** has
a simpler, user-friendly interface that focuses on text editing, formatting and document
management, making it easier for most users to navigate.

(01 Mark @any three differences between DTP and MS Word, total 03 Marks)

(c) The categories of DTP software are:

(i) Page layout and Design Software, eg. Adobe InDesign, QuarkXPress and Scribus.

(ii) Vector Graphics Design Software, eg. Adobe Illustrator, CorelDRAW and Inkscape.

(iii) Raster Image Editing Software, eg. Adobe Photoshop and GIMP.

(iv).Web Design and Development Software, eg. Adobe Dreamweaver and WordPress.

(v) Typographic Software, eg. Front Forge and Glyphs.

(vi). Prepress Software, eg. Adobe Acrobat and Enfocus PitStop Pro.

(01 Mark @any three categories between DTP and MS Word, total 03 Marks)

9. Answer: Any of 5 point @2.5 marks, Any relevant introduction=1.5 marks and conclusion=01 mark.

Running a small business requires handling many tasks, from managing customer records to tracking sales and inventory. If you are still using traditional methods like notebooks or spreadsheets, you might face challenges such as lost records, slow data retrieval, and errors in transactions. A database system can help you run your business more smoothly and efficiently. Here are key benefits of using a database system in your business:

Organize Data Management

A database stores all your business information (customers, products, suppliers, and sales) in a structured way, making it easy to access and update when needed.

Faster Data Retrieval

Instead of searching through piles of papers or multiple spreadsheets, a database allows you to retrieve customer details, sales history, and stock levels instantly with just a few clicks.

Data Accuracy and Consistency

A database eliminates duplicate and inconsistent records. When you update customer or product details, the changes reflect in all relevant sections, reducing errors in transactions.

Better Customer Relationship Management (CRM)

A database helps track customer purchases, preferences, and feedback, enabling you to offer personalized services, discounts, or promotions to loyal customers. This improves customer satisfaction and retention.

Inventory and Stock Control

With a database, you can monitor stock levels in real time, receive alerts when products are running low, and avoid overstocking or running out of essential items.

Increase Business Efficiency

By automating repetitive tasks such as invoicing, order tracking, and sales reports, a database reduces manual work, allowing you to focus on growing your business.

Enhance Security and Data Protection

Unlike paper records that can be lost or stolen, a database has security features like password protection and access control, ensuring only authorized users can view or modify sensitive business information.

Backup and Data Recovery

A database allows you to back up all your important business records, protecting them from accidental loss, theft, or system crashes. In case of any failure, data can be recovered easily.

Improve Decision-Making

A database generates sales reports, customer insights, and financial summaries that help you make informed business decisions, such as identifying best-selling products and adjusting pricing strategies.

Business Scalability and Growth

As your business grows, so does the amount of data you handle. A database can scale up to accommodate more transactions, customers, and employees without affecting performance.

Conclusion

A database system is an essential tool that saves time, reduces errors, enhances security, and improves customer service. It helps your business operate efficiently and compete in today's fast-paced market.

10.Answers

-INTRODUCTION

Definition of computers

-MAIN BODY

In education= Internet, Online learning platforms (e learning), Virtual class rooms

In health care= Improve patient care, Improve medical report, For electronic health records eg x rays ultra sound, Leading to better diagnoses and treatment.

In business= E commerce,

In communication= Social media, e- mails, Video conferencing

-CONLUSION

Student's opinions concerning the question

11. Answer question no 11

Introduction

Any relevant introduction including meaning of presentation.....3 marks

Main body

Any Five characteristics of good quality presentation....2marks@total 10 marks

i.has 6-9 number lines

ii.has 32 readable front size

iii.Brief and readable slides

iv.visual objects must support verbal presentation

v.Do not insert too many images or drawing in one slide

vi.use of color and formatting to emphasize points

Conclusion

any relevant conclusion.....2marks